THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 11: **Nature's Treasures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student collected 10 mL of gas from a jar over water after a candle burnt out. The candle extinguished because: 1

(a) no gas remained **(b)** all nitrogen was used
(c) most oxygen was used **(d)** the jar broke

2. Which of these actions improves soil fertility naturally? 1

(a) adding plastic **(b)** adding compost **(c)** burning plastics **(d)** removing worms

3. A town has 1,00,000 vehicles each emitting smoke. Which step reduces the problem most? 1

(a) less trees **(b)** more public transport
(c) more cars **(d)** more roads only

4. In that experiment, water rises in the jar by about one fifth because: 1

(a) nitrogen forms one-fifth **(b)** water is heavy
(c) jar is wide **(d)** oxygen forms about one-fifth of air

5. Earthworms help soil by: 1

(a) removing humus **(b)** loosening and mixing it
(c) making it hard **(d)** polluting it

6. Which of the following resource-use decisions is sustainable? 1

(a) burning forests **(b)** cutting without planting
(c) wasting water **(d)** planting a tree for every tree cut

7. Which of the following correctly shows the relative abundance in air? 1

(a) carbon dioxide > oxygen > nitrogen **(b)** nitrogen > oxygen > carbon dioxide
(c) nitrogen = oxygen **(d)** oxygen > nitrogen > carbon dioxide

8. When a river dries and the land becomes bare, the soil is more likely to: 1

(a) grow plants **(b)** become fertile
(c) absorb more water **(d)** erode

9. **Assertion (A):** Sunlight is a renewable resource. 1
Reason (R): It is available every day and does not get used up by our use.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Coal is a non-renewable resource. 1
Reason (R): Nitrogen is the most abundant gas in air.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the water rise about one fifth in a jar when a burning candle goes out over water? 2

12. Explain why we say forests are the lungs of the Earth. 2

13. How do earthworms and bacteria keep soil fertile? 2

14. Why does the quality of soil decrease if the same crop is grown again and again? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the oxygen present in a room of air volume 50 m^3, assuming 21 per cent oxygen. 3

16. A school discards 200 sheets of paper daily. If both sides were used, how many sheets would be needed? 3

17. Explain why hilly areas need special care for erosion. 3

OR

A car consumes 1 L of petrol per 15 km. How much petrol will be saved in a month if a person walks or cycles instead of using the car for 60 km? 3

18. Explain why the burning of plastic is worse than burying it in terms of air quality. 3

OR

Describe why wetlands, forests and mangroves are important natural treasures. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A school in Dehradun carried out an activity on air. Students blew into a balloon, burnt a candle under a glass jar and saw it go out after some time, and noticed that moving air turned a small paper windmill. The teacher said air is a mixture of gases, mostly nitrogen and oxygen. 5
(i) Name the most abundant gas in air.
(ii) Which gas is needed for burning and respiration?
(iii) What is moving air called?
(iv) Why did the candle go out in the closed jar?

20. Analyse the case: A town cuts nearby hills for stone, building and farmland. Predict five consequences. 5

OR

Explain how waste from kitchen and garden can be turned into resources. 5

END OF QUESTION PAPER — Total: 40 Marks